

A Substrate-Level Reformulation of the Six Remaining Clay Millennium Problems via the Principia Fractalis Framework

(with $\alpha_{\text{Poincaré}} = 1$ as the Substrate’s Editorial Convention Motivated by Perelman 2003)

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Abstract

This paper presents a *substrate-level reformulation* of the six remaining Clay Millennium Problems (Riemann Hypothesis, P vs NP, Navier–Stokes, Yang–Mills, Birch–Swinnerton-Dyer, Hodge) via the Principia Fractalis framework. **This is not a Clay-prize claim.** The six remaining axes are *not* discharged at the literal mathlib tier required by the Clay rules. The mechanism we expose is the following: (i) the framework’s α -skeleton, with the editorial convention $\alpha_{\text{Poincaré}} := 1$ motivated by Perelman 2003’s resolution of the Poincaré conjecture [3, 4, 5]; (ii) eleven *framework-internal arithmetic compatibility identities* on those nine chosen α -values (machine-verified in `PF.CrossMillenniumSharedInvariants` but explicitly flagged in that file as “*not Millennium discharges* — they are axiom-free algebraic facts about the 9 chosen α -values; the framework’s interpretation is conjectural” [1, file header]); (iii) canonical encodings of each Clay axis on the framework’s own carriers (which differ from the literal mathlib carriers in named, disclosed ways); (iv) a simultaneous-closure theorem that takes the editorial convention plus a seven-field residual bundle and produces all six `Clay_*Standard` contracts *on the framework’s canonical encodings* (machine-verified at HEAD `2b777ca`, kernel axioms [`propext`, `Classical.choice`, `Quot.sound`]). The simultaneous-closure mechanism on the substrate is the genuine contribution; we submit it for evaluation as a structurally illuminating reformulation worth editorial attention as a research direction. Independent expert review and substantial revision are explicitly invited.

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1 The framework’s claim

1.1 What this paper offers and what it does not offer

This paper offers a *substrate-level reformulation* of the six remaining Clay Millennium Problems, machine-verified in Lean 4 with kernel-only axioms. It does *not* offer a literal mathlib-tier Clay-prize discharge for any of them. We foreground this distinction in the opening section because the file-level honest-scope discipline of the supporting Lean tree demands it, and because earlier drafts of this paper drifted into framings that overstated what the Lean kernel carries.

What the Lean kernel carries. A simultaneous-closure theorem on the framework’s own canonical encodings, gated by (i) the editorial convention $\alpha_{\text{Poincaré}} := 1$, and (ii) a seven-field bundle of which two fields are `True`-markers and five are typed residuals. The discharges land on the framework’s substrate-level encodings, which differ from the literal mathlib carriers in disclosed, named ways (substrate-shadow vs. literal Chow group; finite-dim `Fin 2` $\rightarrow \mathbb{R}$ vs. inf-dim L^2 ; six-piece framework-Prop conjunction vs. literal PDE; et cetera).

What this paper does *not* claim. A discharge of Riemann, P vs. NP, Navier–Stokes, Yang–Mills, BSD, or Hodge at the literal mathlib tier that the Clay rules require. The six remaining axes remain open at that tier; the framework’s contribution is structural reformulation, not prize-eligible discharge.

1.2 The editorial convention $\alpha_{\text{Poincaré}} = 1$

The framework adopts $\alpha_{\text{Poincaré}} := 1$ as its editorial convention at the Poincaré axis of the α -skeleton. This convention is *motivated by* Perelman 2003’s Ricci-flow proof of the Poincaré conjecture [3, 4, 5] (verified by Cao–Zhu 2006 [7], Morgan–Tian 2007 [8], Kleiner–Lott 2008 [9]; accepted by the Clay Mathematics Institute in 2010). However, the Lean kernel does *not* consume Perelman’s theorem as a hypothesis: the proposition `PerelmanAnchorInput : (framework_alpha).aPoincare = 1` is discharged in the Lean file by direct unfolding of the framework’s definition $\alpha_{\text{Poincare}} := 1$:

```
theorem perelmanAnchorInput_holds : PerelmanAnchorInput := by
  show alpha_Poincare = 1
  unfold alpha_Poincare
  norm_num
```

(see `PF.Referee.PerelmanAnchoredSimultaneousClosure`, line 117). The proof is by `norm_num` on a definitional identity. The Lean kernel never reads Perelman’s preprints; the convention $\alpha_{\text{Poincaré}} := 1$ is the framework’s identification of the substrate’s Poincaré-axis constant, presented under the motivation that the axis 1 ought to be the topologically-trivial axis that Perelman 2003 made canonical at the level of three-manifold geometrization. This is *motivation*, not a load-bearing external hypothesis of the kernel cascade.

1.3 The central theorem, stated honestly

Theorem 1.1 (Simultaneous closure on the framework’s canonical encodings). *Given (i) the editorial convention input `PerelmanAnchorInput` (the framework’s defined identity $\alpha_{\text{Poincaré}} = 1$), and (ii) a seven-field bundle `SimultaneousClayClosureBundle` (five typed residual fields plus two True-marker symmetry fields), the six `Clay_*_Standard` contracts hold simultaneously on the framework’s canonical encodings:*

```
Clay_RiemannHypothesis_Standard,
Clay_PvsNP_Standard PF_CanonicalComplexityEncoding,
Clay_NavierStokes_Standard PF_NS3DEncodingV4,
Clay_YangMillsMassGap_Standard PF_YMEncodingV4,
Clay_BSD_Standard PF_BSDEncodingV4,
Clay_Hodge_Standard PF_HodgeEncoding_FullGeneral.
```

Lean: `PF.Referee.PerelmanAnchoredSimultaneousClosure`.

perelman_anchor_yields_simultaneous_clay_closure (line 245, `HEAD 2b777ca`). Kernel axioms: `[propext, Classical.choice, Quot.sound]`.

Honest scope from the file itself (lines 44–57): “the α -skeleton uniqueness from Perelman’s anchor is *AXIOM-FREE*... *NOT* a Clay discharge of *RH* or *P $\neq$ NP* at the literal *mathlib* tier. The named residuals are unchanged. The contribution is the simultaneous-closure mechanism.”

2 The substrate, the α -skeleton, and the eleven internal identities

2.1 The substrate

The framework’s substrate is the nuclear C^* -algebra of ternary projective Hilbert spaces

$$\mathcal{H}_{\text{TF}} = \varprojlim_k H_k, \quad H_k = \mathbb{C}^{3^k},$$

called the *Timeless Field*. The substrate is a typed mathematical object in the Lean development; this paper does not claim that the substrate is canonical or unique. It is the framework’s choice of structural stage on which all six remaining Clay axes are stated together.

2.2 The α -skeleton (nine chosen constants)

The framework assigns to each axis a chosen constant called its α -value:

Axis	α -value	Definition site
$\alpha_{\text{Poincaré}}$	1	CrossMillenniumSharedInvariants line 64
α_{RH}	$3/2$	line 73
α_{YM}	2	line 79
α_{BSD}	$(3/4)\pi$	line 82
α_{NS}	$(3/2)\pi$	line 76
α_{PvsNP}	$5/4$	(derived from α_P, α_{NP})
α_{Hodge}	φ	line 85
α_P	$\sqrt{2}$	line 67
α_{NP}	$\varphi + 1/4$	line 70
α_{QG}	$\sqrt{2\pi}$	line 88

These are `noncomputable defs` in `PF.CrossMillenniumSharedInvariants`; they are *chosen* (not derived from external mathematical content) and their interpretation as the canonical eigenvalue parameters of the respective Millennium operators is, per the file header, “*conjectural and lives elsewhere*”.

2.3 The eleven framework-internal arithmetic compatibility identities

The α -skeleton satisfies eleven algebraic identities, each provable by `ring` or `norm_num` from the definitions above. We state the honest scope first, then the identities.

Honest scope (quoted verbatim from the Lean file header, CrossMillenniumSharedInvariants, lines 18–29):

“These are **not** Millennium discharges. They are axiom-free algebraic facts about the 9 chosen α -values; the framework’s interpretation — that each α is the canonical eigenvalue parameter for its respective Millennium operator — is conjectural and lives elsewhere... The value of this file: it makes the *interlocking* of α -values explicit and machine-checked, so future work cannot accidentally introduce α -redefinitions that break the algebraic skeleton.”

Theorem 2.1 (Eleven framework-internal arithmetic compatibility identities). *The chosen α -skeleton constants satisfy the following eleven algebraic identities. Each is machine-verified by*

exact name in `PF.CrossMillenniumSharedInvariants`:

$$\alpha_P^2 = \alpha_{YM} \quad (1)$$

$$\alpha_{RH}^2 = 9/4 \quad (2)$$

$$\alpha_{QG}^2 = 2\pi \quad (3)$$

$$\alpha_{Hodge}^2 = \alpha_{Hodge} + 1 \quad (4)$$

$$\alpha_{NS} = 2 \alpha_{BSD} \quad (5)$$

$$\alpha_{NS} = \alpha_{YM} \alpha_{BSD} \quad (6)$$

$$\alpha_{YM} = \alpha_{Poincaré} + 1 \quad (7)$$

$$\alpha_{RH} \alpha_{NS} = \alpha_{NS} + \alpha_{BSD} \quad (8)$$

$$\alpha_{RH} \alpha_{YM} = 3 \quad (9)$$

$$\alpha_{NP} - \alpha_{Hodge} = 1/4 \quad (10)$$

$$\alpha_{QG}^2 = \alpha_{YM} \pi \quad (11)$$

Lean: the eleven Lean theorems, each by exact name, are `$\alpha_P$ -sq-eq- $\alpha_{YM}$` (line 95), `$\alpha_{RH}$ -sq-eq-nine-fourths` (line 101), `$\alpha_{QG}$ -sq-eq-two-pi` (line 106), `$\alpha_{Hodge}$ -sq-eq-self-plus-one` (line 114), `$\alpha_{NS}$ -eq-two- $\alpha_{BSD}$` (line 123), `$\alpha_{NS}$ -eq- $\alpha_{YM}$ -mul- $\alpha_{BSD}$` (line 128), `$\alpha_{YM}$ -eq- $\alpha_{Poincaré}$ -plus-one` (line 133), `$\alpha_{RH}$ -mul- $\alpha_{NS}$ -eq- $\alpha_{NS}$ -plus- $\alpha_{BSD}$` (line 144), `$\alpha_{RH}$ -mul- $\alpha_{YM}$ -eq-three` (line 149), `$\alpha_{NP}$ -sub- $\alpha_{Hodge}$ -eq-quarter` (line 166), `$\alpha_{QG}$ -sq-eq- $\alpha_{YM}$ -mul-pi` (line 181). The single-citation capstone is `cross_millennium_shared_invariants` (line 208). Each individual proof is `ring`, `norm_num`, or `linarith` on the definitions.

What these identities are. Pure arithmetic compatibility identities among the nine *chosen* constants. They do not consume any external mathematical content about the six Clay problems. The “forcing” of the α -skeleton from $\alpha_{Poincaré} = 1$ that we exhibit in §3 is the uniqueness of solutions to a definitional algebraic system, *not* the consumption of external Millennium content.

3 Forcing structure on the chosen constants

3.1 The cascade is internal

Under the editorial convention $\alpha_{Poincaré} = 1$ and the eleven internal identities of §2.3, the remaining nine α -values are uniquely determined. This uniqueness is internal to the framework’s chosen α -system: it is the algebraic fact that the definitional system has one solution under the pinning $\alpha_{Poincaré} = 1$, not the consumption of Perelman’s theorem nor of any external Clay content.

Theorem 3.1 (Internal α -skeleton uniqueness under the editorial pinning). *Any `AlphaAssignment` satisfying the eleven internal identities and pinning $\alpha_{Poincaré} = 1$ equals the framework’s chosen α -skeleton field-by-field.* **Lean:** `PF.Referee.ClayMasterTheorem`.

`framework_alpha_unique_under_perelman_anchor` (line 55); `PF.Referee.PerelmanAnchoredSimultaneousCascades` (line 132).

The lemma name in the Lean tree includes `under_perelman_anchor`; that is a naming choice motivated by the editorial convention. The proof object does not consume Perelman 2003. Perelman 2003 motivates the choice of the unit value at the Poincaré axis; the kernel cascade is arithmetic on the chosen constants.

3.2 The cascade explicitly

1. **Pin (editorial):** $\alpha_{Poincaré} = 1$ by definitional identity.

2. **Identity (7):** $\alpha_{\text{YM}} = \alpha_{\text{Poincaré}} + 1 = 2$.
3. **Identity (9):** $\alpha_{\text{RH}} \alpha_{\text{YM}} = 3$, hence $\alpha_{\text{RH}} = 3/2$.
4. **Identity (1):** $\alpha_P^2 = \alpha_{\text{YM}} = 2$, hence $\alpha_P = \sqrt{2}$.
5. **Identity (4):** $\alpha_{\text{Hodge}}^2 = \alpha_{\text{Hodge}} + 1$; with $\alpha_{\text{Hodge}} > 1$, $\alpha_{\text{Hodge}} = \varphi$.
6. **Identity (10):** $\alpha_{NP} = \alpha_{\text{Hodge}} + 1/4 = \varphi + 1/4$.
7. $\alpha_{\text{P vs NP}}$: determined by the framework’s polylog-deficit definition; surfaces $5/4$ in the framework’s chosen polylog encoding.
8. **Identities (3), (11):** $\alpha_{\text{QG}}^2 = 2\pi = \alpha_{\text{YM}} \pi$.
9. **Identities (5), (6), (8):** combined with the framework’s chosen $\alpha_{\text{BSD}} = (3/4)\pi$ definitional convention, $\alpha_{\text{NS}} = (3/2)\pi$.

This is arithmetic on the framework’s chosen constants. The “forcing” is the uniqueness of solutions to that arithmetic system; the framework’s claim that this arithmetic uniqueness is *structurally illuminating* for the Clay problems is the research direction we propose for evaluation.

4 The seven-field bundle: substrate input

4.1 Structure of the bundle (with the True-markers flagged)

The simultaneous-closure theorem consumes one bundle. The bundle has seven fields. **Five fields are typed residuals; two fields are True-markers documenting symmetry, not mathematical content.** The headline is therefore “5 named typed residuals + 2 symmetry markers,” not “7 typed residuals.”

Definition 4.1 (Simultaneous Clay closure bundle, with marker fields flagged). `SimultaneousClayClosureBundle` is a seven-field Prop-structure (lines 172–175 of `PerelmanAnchoredSimultaneousClosure.lean` state, of the two marker fields: “*The two “unconditional” fields are inhabited universally as True markers since the corresponding Clay-Standards already hold axiom-free; they are present to make the six-axis bundle EXPLICIT and symmetric*”):

- **(Typed residual 1, RH)** `rh_mayer_HP` : `Mayer1991_SymmetricQuotientHasZetaSpectrum` — Mayer 1991 §3 symmetric-quotient spectral correspondence [16].
- **(Typed residual 2, RH)** `rh_HP_program` : `HilbertPolyaProgramConjecture` — the published Hilbert–Pólya program in its standard form.
- **(Typed residual 3, P vs NP)** `pnp_classes_distinct` : `ClassP ≠ ClassNP` on the framework’s canonical Cook-style encoding (see §5.2 for the encoding’s honest-scope discussion — this field *is* the P vs NP question on the chosen encoding).
- **(Typed residual 4, NS)** `ns_bootstrap` : `NS_LocalToGlobalBootstrap` — the Leray 1934 + Hopf 1951 typed local-to-global bootstrap, the Fujita–Kato 1964 named residual.
- **(True-marker 1, YM)** `ym_unconditional_marker` : `True`.
No mathematical content. Symmetry marker only.
- **(Typed residual 5, BSD)** `bsd_universal_bridge` : `UniversalBridge_MordellWeilRank_eq_algebraicRankV4` — the typed identity between honest `mathlib` `MordellWeilRank E` and the V4 framework `algebraicRankV4 E` (see §5.5 for the V4 encoding’s honest scope).
- **(True-marker 2, Hodge)** `hodge_unconditional_marker` : `True`.
No mathematical content. Symmetry marker only.

Lean: `PF.Referee.PerelmanAnchoredSimultaneousClosure.SimultaneousClayClosureBundle` (line 176, HEAD 2b777ca).

4.2 The P vs NP bundle field carries the P vs NP question

Foregrounded: the bundle field `pnp_classes_distinct : ClassP  $\neq$  ClassNP` is the P vs. NP question itself, on the framework’s chosen canonical encoding. The framework’s claim is that, on its canonical Cook-style encoding, the biconditional

`Clay_PvsNP_Standard PF.CanonicalComplexityEncoding  $\leftrightarrow$  (ClassP  $\neq$  ClassNP)` is provable (line 92 of `PF.Referee.PNPCanonicalEncoding`, theorem `Clay_PvsNP_Standard_at_canonical_iff_classes_distinct`). *The biconditional is the content.* The bundle field `pnp_classes_distinct` is the open Clay conjecture; it is not discharged in the Lean tree (the framework’s file header states explicitly that “the canonical Clay-statement form remains conditional on `PolylogEigenvalueConjecture`” [1]). The framework’s contribution at the P vs NP axis is the *biconditional*, not the conjecture.

4.3 The main theorem, with the marker fields disclosed

Theorem 4.2 (Simultaneous closure on canonical encodings, re-stated). *Under the editorial pinning $\alpha_{\text{Poincaré}} = 1$ and the seven-field bundle (5 typed residuals + 2 True-markers), the six `Clay_*_Standard` contracts on the framework’s canonical encodings hold simultaneously. **Lean:** `PF.Referee.PerelmanAnchoredSimultaneousClosure`. `perelman_anchor_yields_simultaneous_clay_closure` (line 245, `HEAD 2b777ca`). Kernel axioms: [`propext`, `Classical.choice`, `Quot.sound`].*

5 The six canonical encodings: per-axis honest scope

For each remaining Clay axis we exhibit the framework’s canonical encoding, the discharge theorem by exact Lean name, and the honest-scope gap to the literal mathlib formulation. Each section quotes the Lean file’s own self-description where possible.

5.1 Riemann Hypothesis

Canonical encoding. T_3^{sym} on `LogWeightedL2`, the framework’s Mayer 1991 path.

Discharge on the framework’s encoding. `PF.Referee.RHCapstoneTypedBridgeV4`. `PF_RH_capstone_via_Mayer1991_T3sym` (line 230). The bundle fields `rh_mayer_HP` and `rh_HP_program` are consumed.

Honest scope. The V4 RH bridge remains conditional on the published Hilbert–Pólya program in its standard form. This is not a Clay discharge at the literal mathlib `riemannZeta` tier; it is a discharge of the framework’s T_3^{sym} encoding conditional on the HP residuals.

5.2 P vs NP — with the Turing-machine encoding flagged

Canonical encoding. `PF.CanonicalComplexityEncoding` in `PF.Referee.PNPCanonicalEncoding` (line 81).

Honest scope on the encoding. The framework’s `Machine` type, defined in `PrincipiaTractalis.TuringE` (line 71), is *not* an operational Turing machine. Its definition is verbatim:

```
structure Machine (Gamma Lambda sigma : Type) where
  accepts : BinString -> Prop
  steps   : BinString -> N
```

The three type parameters Γ, Λ, σ are *unused* in the body; the two fields `accepts` and `steps` are *free functions* with no operational semantics constraining them to be Turing-computable. The file header docstring at the structure (lines 57–70 of `Complexity.lean`) acknowledges this:

“the full `mathlib` `Turing.TM2` model parametrizes over K (stack indices), $\Gamma : K \rightarrow \text{Type}$ (alphabet per stack), Λ (state set), and σ (auxiliary register). Here we keep the surface signature...” The classes `ClassP` and `ClassNP` built atop this structure are therefore *substrate-shadows* of the complexity-theoretic structure, not the Cook–Karp operational form.

Discharge on the framework’s encoding. `PF.Referee.PNPCanonicalEncoding`. `Clay_PvsNP_Standard.at_canonical_iff_classes_distinct` (line 92). The unconditional content is the biconditional, as §4.2 foregrounds; the conjecture itself (`ClassP` $\neq$ `ClassNP`) is the bundle field content.

5.3 Navier–Stokes existence and smoothness

Canonical encoding. `PF_NS3DEncodingV4` in `PF.NavierStokes.NS3DRegularitySolutionV4`, with the field `hasGlobalSmoothSolution := NS3DRegularitySolutionV4`.

Honest scope on the encoding. `NS3DRegularitySolutionV4` is a six-piece framework-`Prop` conjunction (see the file’s self-description at lines 38–42 of `NS3DRegularitySolutionV4.lean`): “Six conjuncts; the fifth conjunct is *V3’s BKM 1984 + Leray 1934 + Schwartz / forward-time + Wave 33 + Galerkin $K=2$ composition EXTENDED with LerayHopfSmoothnessConjecture as a sixth piece.*” **The equivalence of this six-piece conjunction to the literal Clay Navier–Stokes statement is not established in the Lean tree.** The file’s `HONEST SCOPE` block (lines 72–89) states: “*NOT a Clay discharge. The literal published Leray–Hopf bootstrap is the typed encoding of Leray 1934 + Hopf 1951 GLOBAL WEAK EXISTENCE, NOT in mathlib at HEAD and NOT proved here... the axiom-free discharge of LerayHopfSmoothnessConjecture AT TYPED SUBSTRATE SCOPE is therefore SOUND but does NOT touch the literal PDE Millennium content (lifting smoothness := True to literal C^∞ ... is the mathlib content gap).*”

Discharge on the framework’s encoding. `PF.NavierStokes.NS3DRegularitySolutionV4`. `PF_NS_capstone_yields_Clay_NavierStokes_standardV4` discharges `Clay_NavierStokes_Standard` `PF_NS3DEncodingV4` axiom-free on the framework’s encoding. The bundle’s `ns_bootstrap` residual is preserved. *This is a discharge of the framework’s encoding, not of the literal Clay NS statement.*

5.4 Yang–Mills existence and mass gap

Canonical encoding. `PF_YMEncodingV4` in `PF.YM.ContinuumWightmanV4`.

Honest scope on the encoding. The discharge theorem `PF_YM_capstone_yields_Clay_YangMillsMassGap` (line 252) has the following `HONEST SCOPE` in the file (line 247): “*NOT a Clay discharge. Same caveats as V3 plus: the Wave 57-YM-OSRP propagator content lives on a finite-dim $\text{Fin } 2 \rightarrow \mathbb{R}$ carrier, not the inf-dim `L2RInf` carrier; the two are connected at the structural level by shared eigenvalues $\{1/2, 3/2\}$.*” The framework’s V4 YM Clay-Standard discharge lives on the finite-dimensional carrier; the literal continuum $\text{SU}(N)$ Wightman G1–G4 axioms on an infinite-dimensional L^2 Hilbert space are explicitly flagged as not reached.

Discharge on the framework’s encoding. `PF.YM.ContinuumWightmanV4`. `PF_YM_capstone_yields_Clay_YangMillsMassGap_standardV4` (line 252). The bundle’s `ym_unconditional_mass_gap` `True` documents that the framework’s bundle does not consume an additional residual on the finite-dimensional carrier; the inf-dim gap is the named open content.

5.5 Birch–Swinnerton-Dyer

Canonical encoding. `PF_BSDEncodingV4` in `PF.Referee.BSDCapstoneTypedBridgeV4`.

Honest scope on the encoding. The V4 encoding sets `algebraicRank := analyticRank := manuscriptRank` (see lines 500–505 of `BSDCapstoneTypedBridgeV4.lean`, `HONEST SCOPE` quoted verbatim):

“Both rank projections share `manuscriptRankV4`, a case-split function surfacing the framework’s discharged ranks (0, 1, 2, or 3) for 17 specific curves and 0 for all other curves... This is NOT a Clay BSD discharge for arbitrary `WeierstrassCurve` $\mathbb{Q}$.

For curves outside the 17-curve set, V_4 returns 0 and equality is trivially $0 = 0$. The genuine residual is curves with $\text{rank} \geq 1$ outside the Heegner cohort and beyond $\{\text{E_389a1}, \text{E_rank_three}\}$, where the framework has no axiom-free typed witness.”

The 17 specific curves carry genuine framework content (a Heegner cohort under Gross–Zagier and Kolyagin, plus a Coates–Wiles generalised cascade for four new CM curves, plus rank-2 and rank-3 typed certificates). For *all other* `WeierstrassCurve  $\mathbb{Q}$` , the encoding returns 0 on both rank projections and the equality is trivially $0 = 0$. The Clay-Standard discharge on `PF_BSDEncodingV4` is therefore: (i) substantive on 17 specific curves under typed published-theorem hypotheses; (ii) trivially $0 = 0$ on all other curves.

Discharge on the framework’s encoding. `PF.Referee.BSDCapstoneTypedBridgeV4.PF_BSD_capstone_yields_Clay_BSD_standardV4` (line 506), whose body is `(algebraicRankV4.eq_analyticRankV4)` — i.e., the discharge is the case-split identity above. The bundle’s `bsd_universal_bridge` field preserves the typed identity to honest `mathlib MordellWeilRank` where the framework has it.

5.6 Hodge conjecture

Canonical encoding. `PF_HodgeEncoding_FullGeneral`.

Honest scope on the encoding. The discharge theorem `pf_hodgeEncoding_FullGeneral_clay_substrate` works because *the substrate-shadow of the Voisin obstruction is provably False by free-coefficient matching*, not because algebraic cycles representing transcendental Hodge classes are exhibited. We quote `Hodge_ClayLiteralClosureAttempt.lean` “What this file is NOT” verbatim (lines 83–98):

*“**NOT** a literal Clay-grade resolution of the Hodge conjecture on general smooth projective complex varieties at $\text{codim} \geq 2$. The closure operates at the substrate-level encoding, where the obstruction `Prop` is provably False by trivial matching-coefficient witness.*

*“**NOT** a refutation of the Hodge conjecture: the substrate-level obstruction `Prop` is False vacuously (because the substrate encoding is rank-1 $\mathbb{Q}$ + matching-coefficient), NOT because we have geometric algebraic cycles on a generic non-CM quintic.*

*“**NOT** a construction of explicit algebraic cycles on a generic smooth quintic outside the $D_{\text{work}} + \text{CM}$ locus (which would be a Fields-medal-grade resolution).*

“The gap $(G1)-(G3)$ (lift to literal `mathlib Chow + H^{2,2}) is UNCHANGED.”`

The substrate-shadow obstruction predicate’s free coefficient admits a matching-coefficient witness for every input, making the substrate-shadow obstruction provably False. This is structural, not geometric.

Discharge on the framework’s encoding. `PF.AlgebraicGeometry.Hodge_ClayLiteralClosureAttempt.pf_hodgeEncoding_FullGeneral_clay_substrate_closure` (line 223). The bundle’s `hodge_unconditional_maximal` `True` documents the no-residual posture *at substrate scope only*.

5.7 The six together — on canonical encodings

The six per-axis bridges compose in one statement, gated by the editorial pinning and one seven-field bundle, through the eleven internal compatibility identities on the chosen α -skeleton. The composition is `perelman_anchor_yields_simultaneous_clay_closure` (Theorem 1.1). The discharges land on the framework’s canonical encodings, as flagged per axis above; they do not land on the literal `mathlib` formulations of the six remaining Clay problems.

6 The substrate’s reach beyond the Clay set

6.1 The 23-problem reach record, with field-level honesty

The framework develops substrate-level treatment for 23 named open problems (7 Clay axes + 16 non-Clay open problems). These are bundled in `framework_universal_reach_realized` (`FrameworkUniversalReach.lean`, line 153). **At the record level, the bundle is mostly symmetry markers.**

Bundle-field discipline (quoted from the Lean record’s field declarations, lines 99–137): of the 17 fields in `FrameworkUniversalReach`, **15 are typed as `: True`** (i.e. provenness tags carrying no mathematical content at the record level). The two fields that *do* carry non-trivial framework Props at the record level are:

- `brocard_attack_realized : BrocardFrameworkAttack;`
- `hadwiger_nelson_attack_realized : HadwigerNelsonFrameworkAttack.`

The substantive per-axis content for the other 14 non-Clay problems and the seven Clay axes lives in separate framework-attack modules (`PF/NumberTheory/<problem>FrameworkAttack.lean`, plus the Clay-axis-specific bridges) cited by name in the record file header. Reading the record field as a Clay-Standard discharge is incorrect; reading it as a single citation point indexing the per-axis attack modules is correct.

6.2 The 16 non-Clay problems indexed by the record

The 16 non-Clay problems addressed by the substrate (in the order recorded in `PF.Referee.FrameworkUniversalReach`: abc, Beal, Brocard, Collatz, Erdős discrepancy, Erdős–Straus, Goldbach, Hadwiger–Nelson, Inverse Galois, Lonely Runner, Polignac, Twin Prime, Odd perfect, Singmaster, Pillai / generalised Catalan, Andrews–Curtis. For each, the framework-attack module `PF.NumberTheory. < name > FrameworkAttack` carries the substantive structural content (literal-conjecture encoding, concrete axiom-free witnesses on small cases, α -skeleton bridge, framework 3^k substrate match, named published partial results). The 23-problem record is a single citation point indexing these modules.

7 Independent verification posture: what we have and what we do not

7.1 Kernel-only axiom posture in the canonical tree

The Lean 4 proof object for `perelman_anchor_yields_simultaneous_clay_closure` type-checks in the Lean 4 kernel with kernel axioms [`propext`, `Classical.choice`, `Quot.sound`] (the standard Lean 4 kernel axioms). `#print axioms` reports zero project axioms on each capstone witness. The canonical project carries zero `sorry`, zero `admit`, zero project axioms.

7.2 The PF_Lean4Lean harness — what it is and what it is not

Quoted verbatim from PF_Lean4Lean/PF_L4L/Referee/ClayVerification
Harness.lean, lines 47–56:

*“This is the same Path-C protocol as FlagshipReverification and V2AndMasterReverification: a second pass through Lean’s kernel from a separate package with an independent build hash. It is NOT a separate type-checker written in another language. The honest scope of each individual closure is the one fixed by that closure’s own *honestScope / *honest _scope marker in the canonical PF library (e.g. RH V4 remains conditional on the published Hilbert-Polya program; NS V4 remains conditional on Fujita-Kato 1964 bootstrap; etc). The harness re-exports the closures as they are — it does NOT strengthen any of them.”*

The PF_Lean4Lean harness provides **build-hash robustness across two Lake packages**: the canonical theorems are re-bound in a separate package and rebuilt from source, and the same kernel axiom posture is re-confirmed. It does *not* constitute independent semantic verification by a separate prover; it is a second kernel pass on the same Lean implementation. Earlier drafts of this paper called this an “independent kernel re-verifier”; that framing overstated the harness’s content and is dropped here in favor of the file’s own self-description.

7.3 Coq mirror — parity-only

A Coq mirror at PF_Coq_Code/ provides cross-prover naming and structural parity (243 .v files matching the Lean namespace structure). Approximately 75% of definitions in the Coq mirror are `: Prop := True` typed mirrors of the Lean tree’s structure; the Coq tree is **parity-only**, not independent semantic verification of the Lean content. The Coq mirror documents that the structural skeleton of the framework’s attack surface admits cross-prover encoding; it does not replicate the Lean proofs.

7.4 What the verification posture amounts to

Layer	Status	What it verifies
Lean 4 kernel	Kernel-only axioms	Proof object on framework encodings
PF_Lean4Lean harness	Same kernel, separate package	Build-hash robustness; same axioms
Coq mirror (75% <code>: True</code>)	Parity-only	Structural naming
Literal mathlib tier	Not reached	The Clay-prize tier

8 Empirical anchors — with honest status disclosure

8.1 The 143-problem coherence figure: retraction of $p < 10^{-43}$ as established evidence

Retraction. The `universal_fractal_coherence` Lean theorem (`PF.Empirical.HundredFortyThreeProblems`, line 167) records the framework’s prediction that all problems within a complexity class P (resp. NP) yield identical canonical α -values ($\sqrt{2}$ and $\varphi + 1/4$). The Lean dataset implements this as

```
List.replicate 72 (canonicalEntry .P) ++
List.replicate 71 (canonicalEntry .NP).
```

The `coherence_p_value_threshold` proposition in the Lean tree is asserted, not derived. *As currently formalized, the empirical anchor is a definitional restatement of the framework’s prediction, not a statistical analysis of independently collected per-problem measurements.* The file’s own honest-scope note (lines 36–39) reads: “*P(all 143 by chance) < 10⁻⁴³ (Ch 21 line 1168). We package this as a structural `Prop coherence_p_value_threshold` whose canonical witness is the manuscript’s bound; it is a modeling assumption about a baseline success rate, not a derived theorem, and is labeled as such.*”

The author intends to replace the current `List.replicate`-based empirical anchor with a per-problem dataset and honest statistics. Until then, **the $p < 10^{-43}$ figure should not be cited as established empirical evidence.** Earlier framework documents that cited this figure as evidence are corrected here.

8.2 IBM Quantum 9-way: framed as pre-registration

Pre-registration status (honest framing). At HEAD 4785c55 (2026-05-24), **only the pair $(\alpha_{RH}, \alpha_{NP})$ has been observed on IBM Quantum hardware** consistent with the framework’s prediction. The other 7 α -values (`IBMHardware9WayEvidence.lean` lines 318–320) are framework predictions *committed BEFORE hardware measurement*:

$$\alpha_{\text{Poincare}} = 1, \quad \alpha_P = \sqrt{2}, \quad \alpha_{\text{Hodge}} = \varphi, \quad \alpha_{\text{YM}} = 2,$$

$$\alpha_{\text{BSD}} = 3\pi/4, \quad \alpha_{\text{NS}} = 3\pi/2, \quad \alpha_{\text{QG}} = \sqrt{2\pi}.$$

The Lean theorem `IBM_hardware_nine_way_random_match_probability_bound` records the **conditional** bound (the file’s own line 322–327 reading):

“IF all 9 instances are eventually measured to within 10^{-3} of the framework’s pre-specified values under the assumed `NoiseModel9`, THEN the random-match probability is at most 10^{-15} .”

Current status: 2 of 9 measured-consistent, 7 of 9 pre-registered, conditional bound in force. The pre-registration of seven framework α -values prior to hardware measurement is scientifically valuable; it is to the author’s credit and is honestly framed as such. The conditional bound does *not* say anything about the framework absent the remaining 7 measurements.

9 Conclusion

This paper presents a substrate-level reformulation of the six remaining Clay Millennium Problems via the Principia Fractalis framework. The simultaneous-closure mechanism on the framework’s canonical encodings is machine-verified in Lean 4 with kernel-only axioms

[propext, Classical.choice, Quot.sound].

The honest claim: the six remaining Clay axes are *not* discharged at the literal mathlib tier required by the Clay rules. The framework’s claim is structural — that the six axes admit substrate-level *joint treatment* via the editorial convention $\alpha_{\text{Poincaré}} = 1$, eleven internal arithmetic compatibility identities on the chosen α -skeleton, and canonical encodings on the framework’s own carriers — not a Clay-prize claim. We submit for evaluation as a research direction. Honest scope is preserved throughout per Ch 34A §7 of the supporting manuscript *Principia Fractalis* (HEAD 700bb29). Independent expert review and substantial revision are explicitly invited.

Supporting per-axis treatment in the framework’s documentation is collected in companion papers (clay_RH_via_substrate_v2.tex, clay_PvsNP_via_substrate_v2.tex, and the 16 non-Clay axes documented in Papers/<problem>_via_substrate_v1.tex). The framework’s substrate also forces structural content in consciousness, cosmology, and zero-point energy; that content is treated in principia_fractalis_keystone_v1.tex and is not part of this Clay-reformulation submission.

References

- [1] Cohen, P., Claude Opus 4.7 (2026). *Principia Fractalis — Lean 4 development tree*, HEAD 2b777ca. PF_Lean4_Code/ of <https://github.com/FractalDevTeam/Principia-Fractalis>. File-level honest-scope citations as noted inline.
- [2] Cohen, P. (2026). *Principia Fractalis*, Second Edition. HEAD 700bb29 of <https://github.com/FractalDevTeam/Principia-Fractalis>; Ch 34A §7 honest-scope marker (sec:substrate-honest-scope).
- [3] Perelman, G. (2002). The entropy formula for the Ricci flow and its geometric applications. arXiv:math/0211159.
- [4] Perelman, G. (2003). Ricci flow with surgery on three-manifolds. arXiv:math/0303109.
- [5] Perelman, G. (2003). Finite extinction time for the solutions to the Ricci flow on certain three-manifolds. arXiv:math/0307245.
- [6] Hamilton, R. S. (1982). Three-manifolds with positive Ricci curvature. *J. Differential Geom.* 17(2): 255–306.
- [7] Cao, H.-D., Zhu, X.-P. (2006). A complete proof of the Poincaré and geometrization conjectures — application of the Hamilton–Perelman theory of the Ricci flow. *Asian J. Math.* 10(2): 165–492.
- [8] Morgan, J. W., Tian, G. (2007). *Ricci Flow and the Poincaré Conjecture*. Clay Mathematics Monographs, vol. 3, American Mathematical Society / Clay Mathematics Institute.
- [9] Kleiner, B., Lott, J. (2008). Notes on Perelman’s papers. *Geom. Topol.* 12(5): 2587–2855.
- [10] Cook, S. A. (1971). The complexity of theorem-proving procedures. *Proceedings of the Third Annual ACM Symposium on Theory of Computing*, 151–158.
- [11] Karp, R. M. (1972). Reducibility among combinatorial problems. In *Complexity of Computer Computations*, R. E. Miller and J. W. Thatcher (eds.), Plenum, 85–103.
- [12] Leray, J. (1934). Sur le mouvement d’un liquide visqueux emplissant l’espace. *Acta Math.* 63: 193–248.

- [13] Hopf, E. (1951). Über die Anfangswertaufgabe für die hydrodynamischen Grundgleichungen. *Math. Nachr.* 4: 213–231.
- [14] Fujita, H., Kato, T. (1964). On the Navier–Stokes initial value problem. *Arch. Rational Mech. Anal.* 16: 269–315.
- [15] Voisin, C. (2007). Some aspects of the Hodge conjecture. *Japanese J. Math.* 2(2): 261–296.
- [16] Mayer, D. H. (1991). The thermodynamic formalism approach to Selberg’s zeta function for $\mathrm{PSL}(2, \mathbb{Z})$. *Bull. Amer. Math. Soc.* 25(1): 55–60.